Educational Technology

Principles of Teaching

Professional Education

What is Educational Technology?

Instructional / learning technology

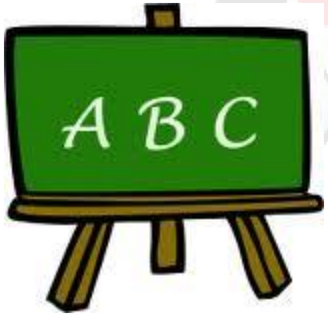
- Theory and practice of design, development, utilization, management and evaluation of processes and resources of learning (Seels & Rickey 1994)
- The study and ethical practice of facilitating learning and improving performance by:
  - creating, using and managing appropriate technological processes and resources

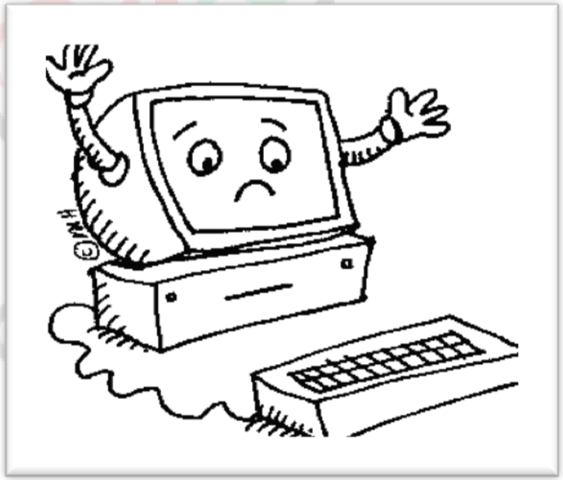
Educational Technology

Encompasses:

Instructional theory : process & systems of learning & instruction

Learning theory : developing human capability





Why Use Instructional Aids?

In GENERAL,

1. Help students remember important information
2. Gain and hold attention of students
3. Useful in supporting a topic
4. Help solve certain language barrier problems
5. Clarify relationship between material objects and concepts (for memory retention and recall)

☺ Why use Educational/ Computer Technology?

In PARTICULAR,

- ✓ Improve motivation
- ✓ Enhance instructional methods
- ✓ Increase productivity
- ✓ Cope with skills requirements of age of knowledge

How technology IMPROVES MOTIVATION

- ✓ Gain learner attention
- ✓ Support for manual
- ✓ Operations in higher learning
- ✓ Illustrations of real world relevance
- ✓ Engagement in production work
- ✓ Connections with distance audiences

How technology
ENHANCES INSTRUCTIONAL METHODS

- ✓ Interaction & immediate feedback
- ✓ Illustrative connections between skills & applications
- ✓ Visual demonstrations
- ✓ Opportunities to study systems in unique ways
- ✓ Unique information sources & populations
- ✓ Self-paced learning
- ✓ Access to learning opportunities
- ✓ Cooperative learning

How technology
INCREASES PRODUCTIVITY

- ✓ Time saving on production tasks
- ✓ Grading and tracking student work
- ✓ Faster access to information sources
  - Distance learning
  - Easy access to learning materials
- ✓ Saving money on consumable materials

Of what use is technology in the KNOWLEDGE AGE

- ✓ Technology literacy
- ✓ Information literacy
- ✓ Visual literacy

Instructor shifts role from "sage on the stage" to "guide on the side"

☹ DISADVANTAGES and LIMITATIONS OF TECHNOLOGY USE

- ✓ Distraction
- ✓ Costly and time consuming
- ✓ Blocks imagination
- ✓ Safety issues

- ☞ Printed page remains best for absorbing dense academic materials
- ☞ Simulations can't replace working on real projects
  - ☞ Knowledge can't be downloaded to the brain
    - 👍 Technology can't replace human element

DOMAINS of EDUCATIONAL TECH

1. **Design** (planning)
2. **Development**
   - **Instructional product development**
3. **Utilization**
   - **Actual use of knowledge and skills by learners**
4. **Evaluation**
   - **of how students learn content under varying instructional conditions**
5. **Management**
   - **links all domains of educational technology together**

INSTRUCTIONAL DESIGN MODELS

1. **ADDIE model**

Analysis

of learning problem, goals, objectives, needs, learning environment, timeline

Design

learning objectives; storyboards, graphic interface, content

Development

production of materials

Implementation

training the teacher and learner

Evaluation

formative and summative evaluation

2. ASSURE Model (Purdue University)

Analyze learners

State objectives

Select media and materials

Utilize media and materials

Require learner participation

Evaluate and revise

Factors Affecting Choice of Instructional Material/Technology

1. Learner
   - preference for learning
   - perception of a given message
   - comprehension of media & language
   - attention span
   - number of learners
   - disabilities
2. Teacher
3. Instructional objective, subject, method
4. Classroom, economic & admin constraints

Checklist on Use of Educational Technology

- ✓**Meaningfulness**

  contain purposive activities, in support of lesson objectives

- ✓**Appropriateness**

  in vocabulary level, difficulty of concepts, methods, interest and developmental stage and age of learners

Checklist on Use of Educational Technology

- ✓**Breadth**

  encompass varied types of learners and satisfies different purposes

- ✓**Usefulness**

  in helping learners learn and teachers deliver content

- ✓**Communication effectiveness**

  relay clear and effective message; properly sequenced

Checklist on Use of Educational Technology

- ✓ **Authenticity**

  encourage student participation which leads to expected behavior of instructional objective

- ✓ **Interest of learners**

  stimulate curiosity and creativity; maintains interest and provides proper stimuli and reinforcement

- ✓ **Allows social interaction**

Checklist on Use of Educational Technology

- ✓ **Cost effectiveness**
- ✓ **Accessibility**

  available to as large and diverse group of learners
- ✓ **Correctness**

  accurate, updated, dependable info
- ✓ **Simplicity**
- ✓ **Assessment**
- ✓ **Portability** (handiness)
- ✓ **Safety**

KINDS OF INSTRUCTIONAL TECHNOLOGY

VISUALS

1. Text/print
2. Drawings, graphics
3. Still visuals (printed/displayed/projected visuals)
4. Realia, models, and dioramas

AUDIO

1. Radio
2. Recorder
3. Cassette

KINDS OF INSTRUCTIONAL TECHNOLOGY

AUDIO-VISUAL

1. Television
2. Cinema
3. Video recordings

COMPUTER-BASED MULTI MEDIA

COMPUTER-MEDIATED MATERIALS

REQUIREMENTS for TECHNOLOGY INTEGRATION

1. **Joint commitment** for technology use among school system/school personnel, teachers
2. **Adequate technology** and curriculum support, hardware, software, and related resources
3. **Well-developed policies** on curricular, legal, and ethical use of technology
4. **Appropriate training** to ensure trained teachers and personnel
5. **Access to technical support** for teachers' and students' computers and software needs

APPLICATION OF ICT in TEACHING and LEARNING

1. **Instructional**
   - Multimedia, computer and internet
   - Computer-Assisted Instruction (CAI) for drills, tutorials, simulation, games, storybooks, problem solving
   - Virtual learning

2. **Informational**
   - Multimedia encyclopedia
   - Teacher and learner sites

APPLICATION OF ICT in TEACHING and LEARNING

3. **Communications**
   - email, chat, teleconferencing
   - electronic bulletin boards, e-whiteboards
   - desktop publishing

4. **Productivity enhancement**
   - Word processing
   - Spreadsheets
   - Database management
   - Presentation

APPLICATION OF ICT in TEACHING and LEARNING

5. Distance education

- **Teachers and learners are physically separated**
- **May be:**
  - **Synchronous –** teacher and students meet *same time but in difference places*
  - **Asynchronous –** *both time and place are different*
  - **Hybrid –** distance education *combined* with face-to-face interaction

EDUCATIONAL ISSUES

- **Standards**
- **Reliance on internet and distance education.**
- **Debate over directed vs inquiry-based, constructivist instructional methods.**

CULTURAL AND EQUITY ISSUES

- Digital divide

  Discrepancy in access to technology among socioeconomic groups
- Racial and gender equity

  Males and whites over females and certain ethnic groups
- Special needs

LEGAL AND ETHICAL ISSUES

- Viruses/hacking
- Plagiarism – cybercheating
- Privacy/safety
- Copyright infringement
- Illegal downloads/piracy

TRENDS in EDUCATIONAL TECHNOLOGY

- **Wireless connectivity**
- **Merging of technologies**
- **Development of portable devices**
- **Availability of high-speed communication (DSL, cable)**
- **Visual immersion systems**
- **Virtual 3-D models**
- **Intelligent applications**
  - **Artificial Intelligence or AI**